CHRIST
(DEEMED TO BE UNIVERSITY)
BANGALORE | DELHI NCR | PUNE

A Data-Centric Platform for Investment Analytics and Financial Market Decision Support

by

Abhijith E (2448503)

Under the guidance of

Dr Thirunavukkarasu V

A Project report submitted in partial fulfillment of the requirements of VI Trimester Master of Science in Artificial Intelligence and Machine Learning

CHRIST (Deemed to be University)

March 2026

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CERTIFICATE

*This is to certify that the report titled **A Data-Centric Platform for Investment Analytics and Financial Market Decision Support** is a bona fide record of work done by Abhijith E (2448503), of CHRIST (Deemed to be University), Bangalore, in partial fulfillment of the requirements of VI Trimester MSAIM during the year 2025-2026.*

Head of the Department | **Project Guide**

Valued-by:

Name & Signature

1. Dr. Kala K.U. Kala 28/3/26

2. [illegible]
28/3/26

Name: ABHIJITH E
Reg. No: 2448503
Examination Centre: CHRIST
(Deemed to be University)
Date of Exam: 28-03-2026

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In-house Internship Offer Letter

Dear Abhijith E,

This is to inform you that, in reference to your interest and further discussion, I am happy to inform you of your selection for an In-house Internship Project on the title A Data-Centric Platform for Investment Analytics and Financial Market Decision Support. The duration of the project will be from 01-01-2026 to 23-03-2026.

The details of the offer are as follows

1. You should report to campus on regular basis from 9 am to 4 pm on all working days. The concerned guide will supervise the project work; the students are encouraged to use library / lab for the project development. On a daily basis, it is mandatory for to maintain the project diary (blue book).
2. The project should be shared with the guide through Github.
3. In general, the interns are not permitted to take leave during the internship period. In emergency cases, prior permission from the department is mandatory to take the leave.

V. Thirukum

Project Guide

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In-house Internship Completion Certificate

This is to certify that Student Name was assigned with the project titled A Data-Centric Platform for Investment Analytics and Financial Market Decision Support from ..01-01-2026... to ..23-03-2026 His performance and competence during the project work is satisfactory.

V. Thangam

Yours Sincerely

Abstract

A Data-Centric Platform for Investment Analytics and Financial Market Decision Support

Financial markets generate vast volumes of data every second, making investment decision-making increasingly complex for retail investors who often lack access to sophisticated analytical tools used by institutional traders. **A Data-Centric Platform for Investment Analytics and Financial Market Decision Support** is designed to address this gap by providing an integrated environment that enables data-driven investment analysis, forecasting, and strategy evaluation.

The proposed platform leverages modern data analytics, machine learning techniques, and scalable system architecture to transform raw financial data into actionable insights. The system enables real-time market monitoring, advanced technical analysis, and predictive modeling for short-term price forecasting using deep learning approaches such as Long Short-Term Memory (LSTM) networks and transformer-based models. In addition, the platform incorporates institutional-grade risk assessment metrics including Value at Risk (VaR), Sharpe Ratio, Alpha–Beta analysis, and drawdown evaluation to help investors understand potential risks and returns.

To support strategy development and validation, the platform provides a comprehensive backtesting and paper trading environment where users can simulate trading strategies using historical market data without financial risk. The system architecture follows a microservices-based design, integrating a modern web interface with dedicated backend services for analytics, portfolio management, and machine learning computation. High-performance databases and caching mechanisms ensure efficient storage and processing of large-scale financial time-series data.

By combining predictive analytics, risk modeling, and strategy simulation within a unified framework, the platform aims to empower investors with professional-grade analytical capabilities. Ultimately, the system serves as a comprehensive decision support tool that enhances investment analysis, improves strategy evaluation, and enables more informed participation in modern financial markets.